

SSS1
'SOUND SYSTEM SIREN'
USER MANUAL

Dub Research, V2, July 2026

Introduction

The SSS1 is a siren FX machine, an extension on the concept of the original NJD siren inna year 3000 style.

Siren sounds can be sculpted, saved, triggered and sent to the built in delay. Whilst traditional siren sounds can easily be sculpted, this device will give you further ability to create new and interesting siren sounds, from soft and delicate, to harsh and destructive.

The siren is unashamedly digital in design, but with some analog bits along the way. I hope this siren will continue to keep moving sound system fwd!

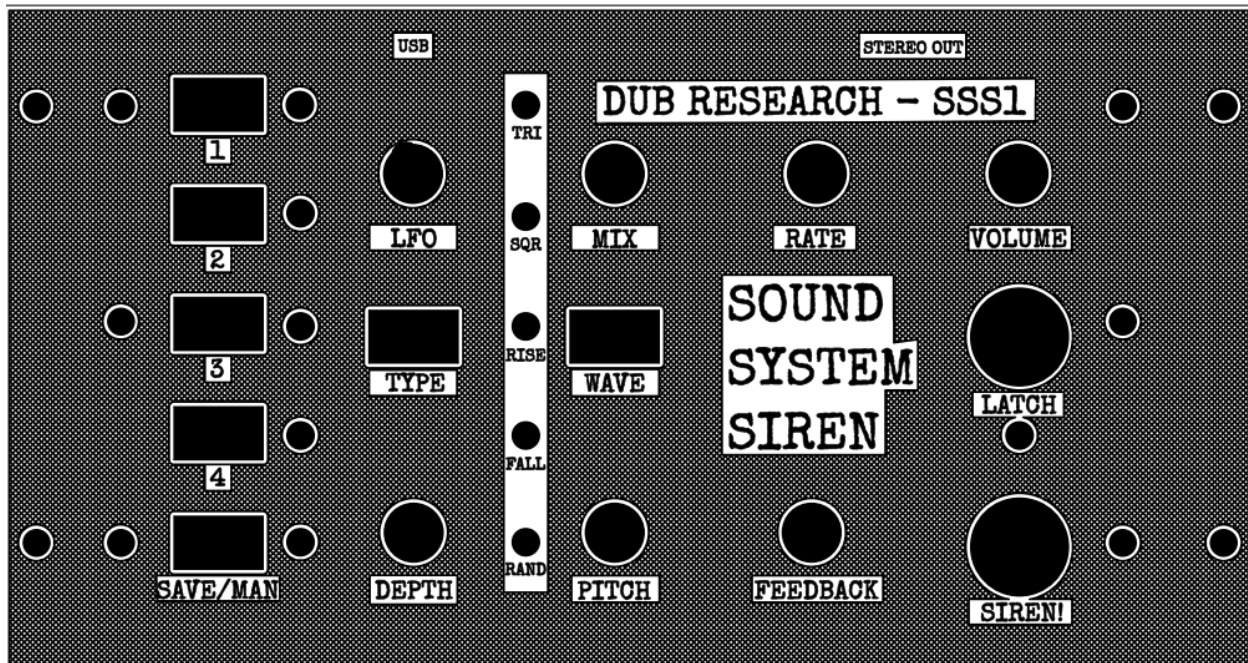


Fig 1 - The Siren

Overview

The siren operates in 4 sections:

1. *The Siren:* Audio comes from a simple 'low frequency oscillator' sometimes referred to as an 'LFO' modulating an audio oscillator. The LFO control controls the rate, or speed of the modulation, depth controls how much, and then pitch controls the overall pitch over the audio oscillator.

The type of LFO and type of audio oscillator can be changed using the respective buttons. Whilst a simple design, it can provide a wide sound palette with very little control. The Siren can be triggered momentarily using the 'SIREN!' button, or toggled on/off using the 'LATCH' button.

2. *Presets:* 4 user presets are available, which save whatever siren you have sculpted. Find the siren sounds you like, save them, easily trigger them later in the dance.

When in manual mode, the current siren sound can easily be saved by holding the 'SAVE/MAN' button and then the preset (1-4) that you would like it stored to, LEDs will indicate that the preset has been saved.

In *Fig 2* below, the example shown is holding the SAVE/MAN button and also holding the preset button 2, to save the current siren to that preset.

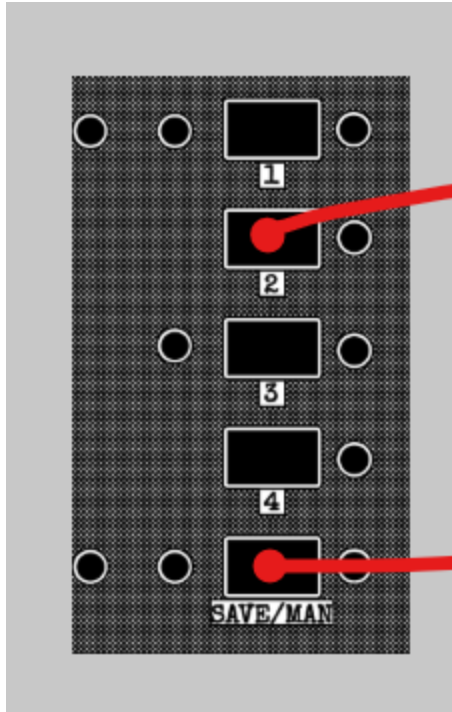


Fig 2 - Saving.

You can jump between presets or switch back to manual mode on the fly, to create complex, evolving fx.

It is worth noting that the memory inside the siren has a lifespan of approximately 100,000 writes. This means you could save 10 presets every day for the next 27 years before the memory starts to deteriorate. Once this happens, the siren will still function and will revert back to the preinstalled presets. I think it is unlikely anyone will reach this point.

3. *Delay:* The dubby lofi delay consists of a dry/wet mix to control the amount of siren/delay.
A rate control, which can go from very fast, BBD style rates, to very long rates (note at longer rates things get a little crunchy, in a nice way).
A feedback control, to get those nice long trailing repeats. With a bit of care on the feedback control, and long rates, near loop style effects can be created.

4. *Output:* The output consists of a single volume control and 3.5mm stereo jack on the rear panel. Therefore, output is in stereo. 3.5mm was chosen as 3.5mm to RCA cables are common and cheap, easy to plug in to mixers, audio gear, even portable speakers (using a 3.5mm to 3.5mm cable) so you can annoy people in the park.

It is worth noting that the siren is powered from a micro-usb connector, it doesn't draw much current and can be powered from a phone charger, battery, even the little usb ports on the bus.

Easter Egg Modes

The siren contains 2 hidden, special easter egg modes:

- Space Drum/Syndrum mode.
- FM drum Synth mode.

To access these hidden modes, simply hold down 'type' + 'wave' and then a preset button to select the mode

1. Siren mode.
2. Space Drum Mode
3. FM Drum Mode.

Space Drum Mode

The space drum contains an oscillator, filter and envelope to make syndrum-esque drum tones. There is also a white noise oscillator included in that mode, to create white noise crash sweeps.

It is possible to save presets in space drum mode, these can be accessed whilst either in Siren or Space Drum modes.

The latch button will act as a small looper for the space drum sound. Whilst in preset mode, the rate knob controls the rate of that looping.

FM Drum Mode

The FM drum synth is a voyage of your own discovery....

Firmware Update Procedure:

To perform the update you will require:

- A Mac, Windows or Linux based computer,
- An internet connection,
- A bottle of dragon stout,
- A data capable micro-usb cable (some cables only provide power for charging).

With power turned off. Hold down the following 3 buttons simultaneously:

- Preset 2
- Preset 3
- Latch

and then connect the usb cable from the siren to your computer. Once this is done, a drive will appear on your computer, just like a usb thumb drive.

Firmware update files can be downloaded from our website and are available in the 'Downloads' section of the SSS1 part of the site.

<https://dubresearch.github.io/>

If the drive is named 'RPI-RP2' then use the download:

Download Firmware v2 (RPI)

https://dubresearch.github.io/sss1_v2_firmware_RPI.uf2

If the drive is named 'RP2350' then use the download:

Download Firmware v2 (RP2350)

https://dubresearch.github.io/sss1_v2_firmware_RP2350.uf2

Later models of the siren use a slightly different processor than the previous ones. So make sure you use the correct file!

Simply drag the file onto the drive that has appeared on your computer. Once that is done, the siren will automatically restart and the drive on your computer will disappear. Congratulations, you just updated your SSS1 Sound System Siren!

Firmware Update Notes:

V2 Firmware Update 29/07/2026

- Upper range of pitch knob reduced.
- An additional, classic style square wave (there are now 5 waveforms)
- LFO can be re-triggered by holding down '*type*' whilst pressing '*save/man*'. This re-trigger is off by default and can be saved to a preset.
- Switch between easter-egg modes without needing to power on/off. Hold down '*type*' + '*wave*' and then a preset button to select the mode, the following preset buttons access these modes:
 1. Siren mode.
 2. Space Drum Mode
 3. FM Drum Mode.
- Space Drum sounds can be saved as presets.
- Both Space Drum and Siren presets can be accessed in either of their respective modes.